Cooperative Multi-Agent Generative Jamming of UAV Networks under Detection Constraints

Bachelor's Thesis Project Proposal

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1 Introduction

Unmanned aerial vehicle (UAV) networks are increasingly deployed as relays, sensor swarms and mobile ad-hoc infrastructure, and jamming remains their dominant physical-layer threat [1]. The countermeasure literature has converged on learned, frequently multi-agent responses: repositioning relay UAVs away from an interferer, coordinated spectrum access and frequency hopping, and joint trajectory and power adaptation [2, 3]. These methods are effective, and share a structural property that is rarely examined: they are *reactive*. Each is triggered by, or conditioned on, an observation that jamming is present — a detector output, a classified jammer type, or an observed collapse in link quality serving the same purpose.

That trigger is treated as reliable because the detection literature reports it as solved: convolutional classifiers on spectrograms of the received signal achieve jamming detection accuracies above 99 % in exactly the OFDM/UAV setting these defenses assume [4]. The resulting pipeline — detect, classify, then relocate, hop or re-power — inherits a single point of failure. If an attacker degrades the link while remaining statistically indistinguishable from the un-jammed condition, the countermeasure is never invoked: the defense does not degrade gracefully, it simply never activates, and the network experiences the degradation as ordinary channel impairment. The exceptions are *proactive* schemes that do not condition on detection at all — uncoordinated frequency hopping [5] and proactive data-driven relocation [6] being the clearest — which makes them the most informative comparison points, since they should retain performance where reactive schemes lose it entirely.

Whether such an attack is constructible is an open question, and this thesis treats it as one. Effectiveness and detectability are coupled by physics — interference that flips bits necessarily perturbs the received constellation — but loosely enough to leave design margin, since interference displacing symbols *across* their decision boundaries rather than along them causes the same errors at lower energy, and lower energy is harder to detect. Two choices exploit that margin. *Generative* policies parameterize the distribution of the transmitted interference directly [8, 9] instead of selecting from a discrete action set, which is what allows a signature to be shaped rather than a channel merely chosen — the mechanism underlying generative covert communication [10]. *Multi-agent* coordination is used because splitting power and phase across several jammers so their contributions add at the victim while each stays individually below threshold has no closed-form solution, and because cooperative reinforcement learning already works for the jamming task itself [7]. The result is an evasion attack against a deployed classifier under constraints the adversarial machine learning literature does not impose [11]: the perturbation must be physically realizable through the attacker's own channel, and must raise the error rate rather than merely flip a label.

2 Research Questions

RQ1. Can a cooperative multi-agent, generative jamming policy degrade a UAV link while remaining below the detection threshold of a state-of-the-art learned detector, and what effectiveness–detectability trade-off does it achieve relative to baselines? The baselines span the full range of attacker capability: barrage interference, a closed-form minimum-energy attack, a single-agent learned attacker, and an omniscient cancellation attacker as an upper bound.

RQ2. How much of the reactive anti-jamming stack still functions under such an attack? This is measured at the level of the *countermeasure* rather than the detector — whether the trigger fires, how long it takes, and what network-level performance results — against proactive, detection-free defenses as the comparison.

RQ3. What does adaptation cost either side? Treating the interaction as a round-based offline game — frozen detector, attacker optimized against it, detector retrained on those attacks, attacker re-optimized — this asks what re-closing the gap costs the defender in accuracy and false-alarm rate, and how much of the loss the attacker subsequently recovers.

3 Methodology

Minimal model first. The work starts from the smallest model in which these questions remain meaningful and adds one layer at a time, each addition justified by a question the previous layer could not answer — not for simplicity's sake, but because in a small model the defender's optimal decision rule is analytically available, which upgrades "this trained detector fails to see the attack" into the far stronger "no detector can do better". The base model is a single QPSK link over an additive white Gaussian noise channel attacked by one interferer under a hard power budget, with detection at the receiver from the received signal alone. Noise is swept starting from the noiseless case: without noise the un-jammed constellation is a finite set of exact points and any perturbation is certainly detectable, so the stealth region exists only as a consequence of noise. Extensions follow in order: several coordinating jammers with distinct link gains and phases; imperfect attacker synchronization, which weakens the attacker and strengthens any surviving result; a multicarrier waveform over a frequency-selective fading channel; and finally UAV mobility and network-level evaluation.

Attacker. The attack policy is generative and trained under centralized training with decentralized execution [12] through a multi-agent-native environment interface [13]. Its objective is the induced error rate penalized by detection probability, with the transmit-power budget imposed as a hard environment constraint rather than a reward term. The training/execution asymmetry is stated explicitly as a modelling assumption: the error rate is available to a centralized critic during training, whereas a deployed jammer observes only its own transmission, its own channel estimate, and at most an acknowledgement signal from the victim.

Defenders and baselines. The defender is a suite rather than a single model: an energy detector at fixed false-alarm rate, a learned spectrogram classifier following the current state of the art [4], and — where the model admits it — the optimal likelihood-ratio test as reference. Above the detector sits the countermeasure itself, so that RQ2 can be answered: a reactive, detection-triggered response (relocation or spectrum reallocation [2, 3]) and a proactive, detection-free one [6, 5], evaluated side by side under the same attack.

Evaluation. Attack performance is reported as bit and symbol error rate against detection probability at a fixed false-alarm rate, under two conventions: strategies are compared at *matched detectability* rather than at equal transmit power, since an attack that raises the error rate by raising its own signature is not an improvement; and every figure carries the full reference envelope from the no-attacker floor to the omniscient ceiling. Defense performance is reported as trigger rate, time-to-trigger and network throughput. Parameter studies cover noise level, power budget, the number of jammers and legitimate nodes, and the exploitable structure in the transmitted sequence. Experiments use Sionna [14] for the physical layer and PyTorch for the learned components.

Risks. The principal risk is that the learned attacker does not separate from the closed-form minimum-energy attack on a single link — plausible, since against an independent and uniformly distributed payload that perturbation is available analytically and a learner can at best rediscover it. This is anticipated rather than avoided: the corresponding ablation is scheduled early, and a negative result redirects the work toward the coordinated multi-jammer setting, where no closed form exists and which is in any case the more relevant threat model for UAV swarms. A second risk — results that look strong only because the attacker was granted unrealistic knowledge — is answered by the assumption ladder, in which capabilities are removed in defined steps.

Work plan. **(1)** System, defender and threat model specification. **(2)** Minimal model: link, attacker ladder, detector suite, optimal-test reference. **(3)** Effectiveness–detectability characterization against the baselines (RQ1). **(4)** Generative and multi-agent attacker; comparison at matched detectability (RQ1). **(5)** Countermeasure-level evaluation, reactive versus proactive (RQ2). **(6)** Adaptation-cost rounds and synchronization realism (RQ3). **(7)** Multicarrier, fading and mobility; writing. Phases 1–3 are prerequisites; 5–7 are ordered by expected value and may be truncated from the end.

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